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MANUSCRIPT BACKGROUND

The work in this thesis was presented as a poster at the Symposium on Fusion Engineering, held 23–26 June 2025 on the MIT Campus in Cambridge, MA. This thesis is an extended version of a six-page conference paper that is being submitted to the IEEE Transactions on Plasma Science journal for the SOFE 2025 special issue. The formatting of this thesis is extremely close to the IEEE requirements (including the required biographies and Americanized spelling), but with some basic changes like the inclusion of an appendix (to house what would otherwise live on the journal's website as supporting material) and the removal of references to submission and publication dates. Also please note that these pages are significantly denser than the 20 page (single column) limit, so the length of the paper is proportionally shorter.

ORIGINALITY STATEMENT

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2D Hall Arrays for High Resolution Tokamak Magnetic Field Imaging

Samuel Shelton and Will Midgley

Abstract—**This paper presents the use of a two-dimensional array of magnetic field sensors designed to image the magnetic field produced in the vacuum vessel of the upcoming SOUTH tokamak. Designed to de-risk engineering efforts by allowing for precise mapping of toroidal and null field configurations within the plasma boundary region, this represents the first use of such a device within the context of fusion. Using low-noise circuitry and multiple analogue-to-digital converters in parallel, the device can achieve up to 20 kHz of analogue bandwidth across 256 sensors, with an effective sensitivity of 0.32 mT against a dynamic range exceeding ±150 mT. The device can also be operated in a real-time mode using a micro-controller, allowing interactive visualizations of magnetic fields for outreach and pedagogical purposes. Preliminary imagery and simulated reconstructions are presented, with the device intended to assist with the research and development, manufacturing and commissioning of the SOUTH tokamak in anticipation of a first pulse in late 2026.**

Index Terms—**tokamak, diagnostics, instrumentation, magneto-vision, Hall effect, magnetic field mapping**

I. Introduction

THE SOUTH tokamak (shown in Fig. 1) is being developed at the University of New South Wales in Sydney, Australia as a student project called *AtomCraft* [1]. SOUTH—the Student Operated Undergraduate Tokamak with Hydrogen—will be the first tokamak designed and built entirely by undergraduates. It will be a compact, spherical tokamak employing conventional magnets and utilizing a combination of ohmic heating and electron cyclotron resonance heating (ECRH). The design is influenced primarily by the Danish NOTH tokamak [2] and its predecessor, Tokamak Energy's ST-25 [3], with the coil and power supply designs inspired by the University of Seville's SMART [4].

SOUTH is expected to operate with a toroidal on-axis magnetic field of 100 mT and an approximate 100 ms pulse, although these specifications are targets and are subject to change. The device has a major radius of approximately 0.3 m, with a vacuum vessel diameter of 30 cm. Initial plasma operations are expected to begin in late 2026, with plasma currents of 3 kA and plasma temperatures of about 10 eV expected [1]. *AtomCraft*'s primary focus is pedagogical, with the initiative focused on providing students with hands-on experience in fusion engineering from as early as their second undergraduate year. SOUTH is *AtomCraft*'s first project, with more nuclear-related projects and upgrades planned for the future.

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Supporting materials for this paper are available at https://github.com/Sam-js2/BEThesis and https://youtu.be/sB7mTB_tMe0

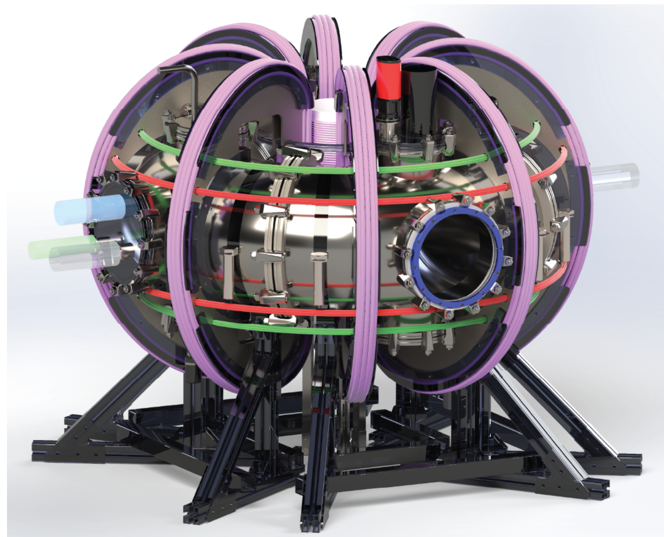

Fig. 1. A render of the upcoming SOUTH tokamak, with placeholder instrumentation, vacuum, fueling and RF equipment, with the 'Princeton Dee' toroidal field (TF) coils are in purple and the poloidal field (PF) coils in red and green [1].

As a student project, *AtomCraft* operates under strict budgetary constraints and with a strong operational emphasis on safety and risk management. A significant portion of funding is sourced externally from philanthropic institutions and commercial fusion organizations, necessitating clear and communicable results for outreach purposes. The idea of a two-dimensional Hall array emerged as a tool to detect technical issues with the magnet system (particularly the critical toroidal field, produce by the 'Princeton Dee' magnets with a configuration as shown in Fig. 2) prior to plasma operation, with potential as an outreach and recruitment tool due to its visual and interactive nature.

A. Hall sensors and magnetic sensor arrays in Fusion

Hall effect sensors are widely used in fusion engineering for magnetic field monitoring [5], plasma diagnostics [6], and current sensing [7]. Low-density arrays have an established use in magnetic field mapping [8], and a high-density one-dimensional array of Hall sensors has been demonstrated as an effective tool for measuring both vacuum and plasma-generated poloidal magnetic fields in magnetically confined steady-state plasmas [9].

1) ITER Magnetic Diagnostics: Chavan *et al.*'s 2009 paper in *Fusion Engineering and Design* serves as a summary of all the magnetic field diagnostics intended for use at ITER, while

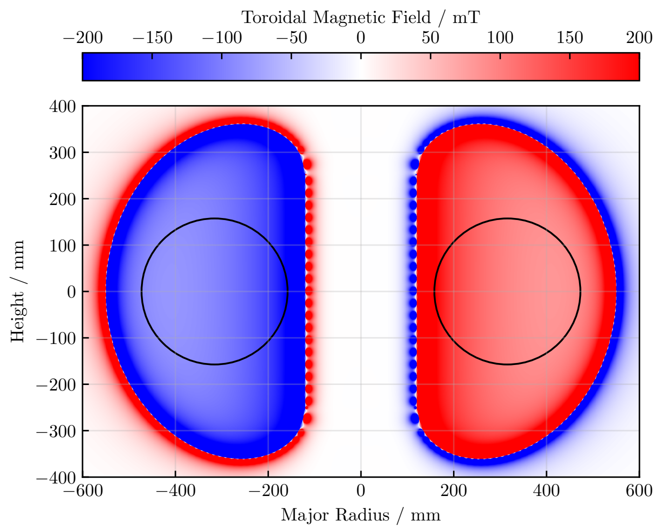

Fig. 2. Toroidal magnetic field generated by SOUTH during a 3 kA plasma pulse. The field here is clipped to ±200 mT and was computed using the Biot-Savart law on a discretized model of the ‘Princeton Dee’ TF coils [1].

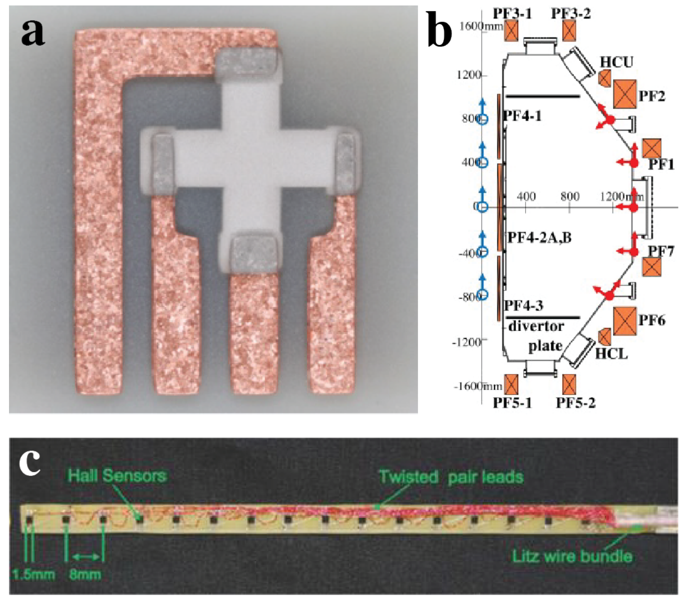

Fig. 3. Fusion applications of Hall sensors: a) Custom Lead-Bismuth Hall sensors for ITER [10]; b) Hall sensor placement for QUEST [5]; c) Insertable Hall array for poloidal field monitoring [9].

also discussing risks to the project [6]. The diagnostics set of ITER is to consist of 120 Hall probes for tangential field monitoring, 54 for radial field monitoring and 12 for toroidal field monitoring. This data is to be used to inform closed-loop control algorithms, and in post-pulse plasma reconstruction. These Hall probes are heavily redundant to account for any probe failures due to excess exposure to radiation [6]. Chavan *et al.* further identify the ITER-specific design drivers that informed the selection of this diagnostics setup. They include the ability to work over ITER’s long plasma pulse lengths, in the presence of high levels of neutron radiation, steep requirements for accuracy, precision and system availability, resistance to various kinds of electrical interference, and several requirements regarding data acquisition [6].

In a related follow-up paper, Ďuran *et al.* detail the design of a custom semiconductor Hall Effect sensor for use at ITER [11]. To address the challenges previously identified by Chavan *et al.* [6], Ďuran *et al.* use a metallic bismuth process and large sensor area to achieve sufficient radiation hardness. Their custom sensor, shown in Fig. 3a, is able to operate with sufficient sensitivities across a large range of magnetic field strengths up to ±6 T, with radiation hardness well in excess of anything commercially available. Fortunately, neutron irradiation and extended pulse lengths are not anticipated at SOUTH, and there is no closed loop control planned for the initial operating configuration. While ITER’s diagnostics are cutting edge, they are not as applicable to UNSW’s much smaller tokamak as might initially be expected.

2) Magnetic Field Diagnostics in Spherical Tokamaks: Other, spherical Tokamaks are more comparable to SOUTH geometrically. The QUEST Tokamak, a mid-sized spherical Tokamak located in Japan, uses an advanced closed-loop control system to achieve extremely stable plasma pulses lasting up to six hours [12]. A key part of this control system is a complex arrangement of Hall effect sensors (both single and tri-axial) located at key points on the outside of the vacuum vessel, as shown in Fig. 3b. These sensors provide output data which is integrated at 100 kHz, and fed into a control loop running at 4 kHz [5]. The sensors themselves are accurate over a range of ±1600 G (160 mT), which is comparable with the peak field expected on SOUTH. The Hall sensors here are particularly useful for combating the drift accumulated by the analog integration of Rogowski coil signals over long plasma pulses [5].

The small CASTOR tokamak featured a small array of eight commercial Hall effect sensor ICs integrated into the stainless steel vacuum vessel [10]. Ďuran *et al.*—the same Ďuran from ITER [11]—integrated sensors, power and signal conditioning circuitry onto a single small PCB, and put the entire device inside the Tokamak vacuum vessel—the first ‘in-vessel environment’ use of such sensors. The results prove the efficacy of commercial Hall effect sensors in environments with relatively low (compared to ITER) amounts of radiation and ambient temperatures up to 150 °C [10]. These results are very relevant to the SOUTH tokamak, which is expected to have conditions (in terms of radiation and ambient temperatures) similar to CASTOR. While Ďuran *et al.*’s work is not directly applicable to SOUTH because it is designed as a plasma diagnostic, the success of the project proves the viability of using commercial sensors for fusion applications.

3) Insertable Linear Array Probe: Stevenson *et al.*’s 2014 paper presents a linear array of 16 Hall effect sensors with a spacing of 8 mm, designed and fabricated on a printed circuit board [9]. This linear array (essentially a long, thin stick, as shown in Fig. 3c) is then inserted into the plasma through a hole in the vacuum vessel, and rotated around its axis of insertion to capture a full range of data. This implementation is extremely innovative and the most comparable existing design to what is proposed in this paper, and the use of twisted pairs, differential amplifiers and analogue anti-aliasing filters distinguish it as

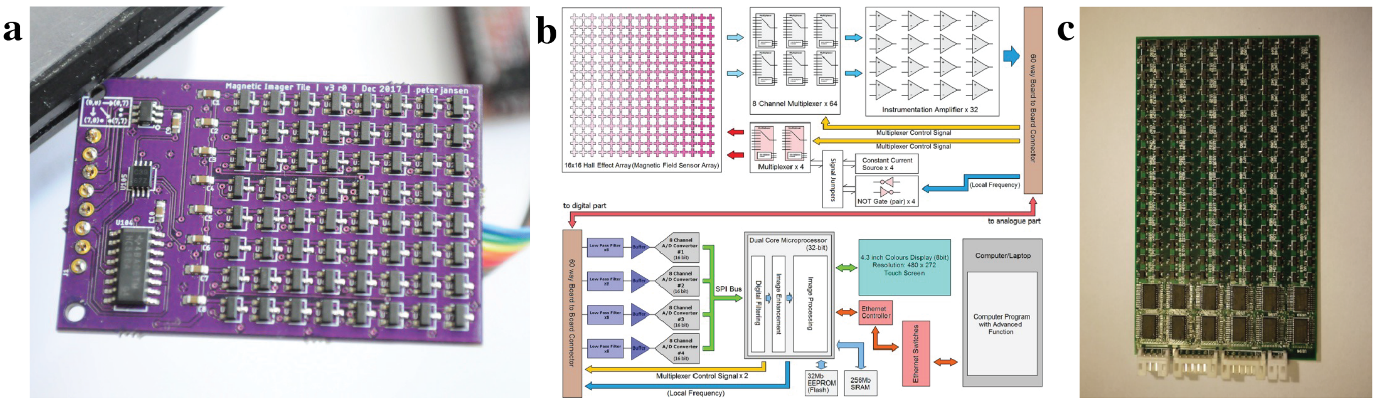

Fig. 4. Non-fusion applications of Hall arrays: a) Jansen's Tile V3 [13]; b) Block diagram of Liang's implementation [17]; c) *Gaussense* PCB [14].

the most complete implementation thus far.

B. Hall Arrays Outside of Fusion

Two-dimensional Hall arrays have been used outside of fusion as educational tools [13], in consumer electronics [14], and for non-destructive testing [17] and metallic object detection [18]. These devices all share a basic systems architecture underpinned by the use of multiplexed sensors, but implementations vary widely in terms of the data acquisition architecture and varying levels of analog filtering and complexity.

1) The Works of Peter Jansen: Peter Jansen's work on these sensors on *Hackaday* served as a strong initial inspiration for this project. Jansen's first Hall board was a 12×12 array that featured closely packed Hall Effect ICs fed into an analog-to-digital converter (ADC) through cascaded I2C multiplexers. This implementation was slow—only achieving 10–30 samples per second (SPS) operation depending on the microcontroller used—mostly due to the speed of the I2C bus [15]. His next iteration, version 3.0 (shown in Fig. 4a), used a smaller 8×8 array and a binary counter connected to a clock to operate the multiplexers, enabling higher sampling rates up to 2,000 SPS [13].

While a great improvement, Janesen notes that the implementation is still bottlenecked by the slow ADC [13], and future devices could get around this by using a more powerful microcontroller. All of Jansen's tiles suffer from a notable lack of any form of buffering or filtering, leaving the digitized signal to be susceptible to aliasing and rendering the entire setup vulnerable to electromagnetic interference. The PCB is also designed without a ground plane, among other issues that make the design susceptible to both the creation and absorption of EMI, necessitating a complete analog overhaul if the device were ever to be applied to fusion.

2) Real-time Magnetovision Camera: Liang *et al.*'s 2017 paper details a well designed and tightly integrated magnetovision imaging device [17]. Their device features a 16×16 array of custom Hall effect sensors whose outputs are multiplexed, buffered and sampled with an ADC—four in parallel for the 16×16 array—commanded by a microcontroller, which also coordinates the operation of the custom Hall effect sensors, as shown in Fig. 3b. Liang *et al.* apply best practice analog design using anit-aliasing filters and buffers, and completely separate the digital and analog PCBs to provide a level of modularity that makes manufacturing and testing easier, while also allowing for the digital components to be readily swapped out.

Liang *et al.* ran their SPI busses at 21 MHz and stored the excessive amount of data generated in discrete external memory, before sending it to a PC for analysis over ethernet [17]. Although designed to operate in 'real-time,' Liang *et al.* still achieve an impressive 600 SPS, more than enough for their specific use case but not enough for imaging the fast changing fields on SOUTH. This real-time application is also why a bilinear approach is used over the preferable but much more computationally intensive bicubic method for spatial interpolation. The design itself is also much to bulky to fit inside SOUTH's vacuum vessel, but the innovative use of separate modular analog and digital PCBs could provide a path to compact and performant implementation.

3) Hall Arrays for User Interface Design: Liang *et al.*[1] proposed the use of an array of commercial Hall effect sensors as a way to sense the orientation of a writing stylus for a digital tablet display in their 2012 paper [14]. This *Guassense* technology was designed to enable devices to distinguish between stylus and finger inputs, sense the tilt and angle of a stylus, and detect a stylus hovering over the surface of a screen. *Guassense* was designed to augment existing devices, being placed behind LCD screens to add a new layer of user interactivity.

The hardware implementation, shown in Fig. 4c, contained 192 commercial Hall sensors multiplexed into a microcontroller's inbuilt ADC. Bicubic interpolation was used to enhance the spatial resolution of images, and the polling rate was "consistently above 60 fps" [14]. *Guassense* suffers from most the same issues as Jansen's implementations, and could benefit greatly from the incorporation of a proper buffering and filtering regime. It is worth commending the vision of *Guassense*: since its publication in 2012, Apple has gone on to successfully commercialize many of the unique goals of the project, namely the detection of stylus angle, tilt and hover [16]. While Apple doesn't use an array of Hall effect sensors, Liang *et al.* successfully anticipated a market demand that

[1]A different Liang *et al.*. from [17].

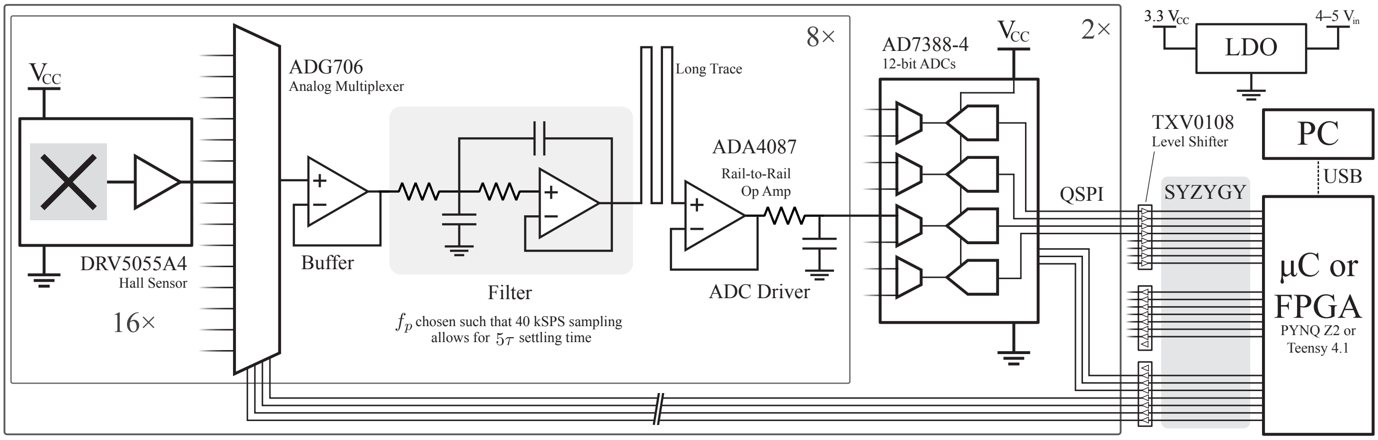

Fig. 5. Simplified circuit schematic showing the operation of the device.

would be only be realized more than a decade later.

4) Unpackaged Scanned Array: Cai *et al.*'s 2024 paper presents an array of 256 unpackaged (not contained in black plastic packages) Hall effect sensors that moved around on an X–Y drive to image a large area at an extremely high resolution [18]. Cai *et al.* are primarily concerned with spatial fidelity, so the use of unpackaged sensors allow the ‘pixels’ to be placed extremely close together, and the X–Y drive allows them to combine adjacent images to enhance both resolution and frame size. Cai *et al.* use external data acquisition boards, power supplies and control systems, so while the sensor itself is compact, the system still has a large overall footprint. While Cai *et al.*'s work is likely the least applicable to fusion applications of the devices discussed due to its extremely slow sampling rate, the use of unpackaged sensors has potential for future, more advanced devices.

C. Tokamak-specific Requirements for Hall Sensors

Based on existing implementations of Hall arrays both inside and outside of fusion and identified design constraints, the following were identified as the key properties required of a Hall array for use in magnetic field imaging on the SOUTH tokamak:

Sensitivity and Range: The array must measure both the strong toroidal field at approximately 100 mT and potential near-zero stray radial magnetic field components; an effective sensitivity of ≤ 1 mT is highly desirable.

Bandwidth and Sampling: Given the short pulse duration of 100 ms, the array requires a high analog bandwidth and associated sampling rate. A bandwidth exceeding 10 kHz—the PWM switching speed of SOUTH's power supplies—is desirable.

Miniaturization: The array must fit within the vacuum vessels and through its standard flanges with a maximum width of 150 mm, while still covering the plasma boundary region of interest.

Environmental: The array must maintain a low noise floor (in order to achieve the desired sensitivity) and resist environmental sources of electromagnetic interference (EMI), including switching transients and RF interference, particularly around the industrial, scientific and medical (ISM) band at 13.56 MHz.

Existing magnetic field diagnostics in Tokamaks exist largely in service of physics applications, rather than in verifying the engineering itself and the ability to construct the magnetic field conditions intended. This is largely due to the much higher degree of engineering confidence in large scale projects like ITER, especially when compared to undergraduate work at UNSW. What diagnostics do exist are largely external to the vacuum vessel itself and are predominantly coil based, sacrificing almost all spatial resolution for temporal resolution and bandwidth.

II. Methodology & Device Description

The final device works by simultaneously sampling eight sensors using two discrete ADCs, with 16:1 multiplexing into each filter and additional 2:1 multiplexing in-package on the ADC ICs; the device operating methodology summarised in Fig. 5. The analog and digital electronics are physically separated, allowing the analog sensor board to fit inside the vacuum vessel while the digital control board is located externally. This modular design also supports multiple digital processing configurations, customized for specific use cases. A 1/16th scale prototype was constructed to de-risk the final design, which was manufactured and assembled by a PCB assemblyhouse in China that used a combination of standard and specialized imported components.

A. Analog Electronics

The selected Hall sensor is the Texas Instruments DRV5055A4, which supports a measurement range up to ± 179 mT when powered at 3.3 V and offers an analog bandwidth of 20 kHz [19]. Each sensor outputs a voltage proportional to the magnetic field ($V \propto B$), with zero field corresponding to $V_{\mathrm{CC}}/2$. This architecture enables all 256 sensors to share a common reference with both ADCs, ensures that all voltages remain within the linear region of all operational op-amps, and provides some common-mode immunity from EMI.

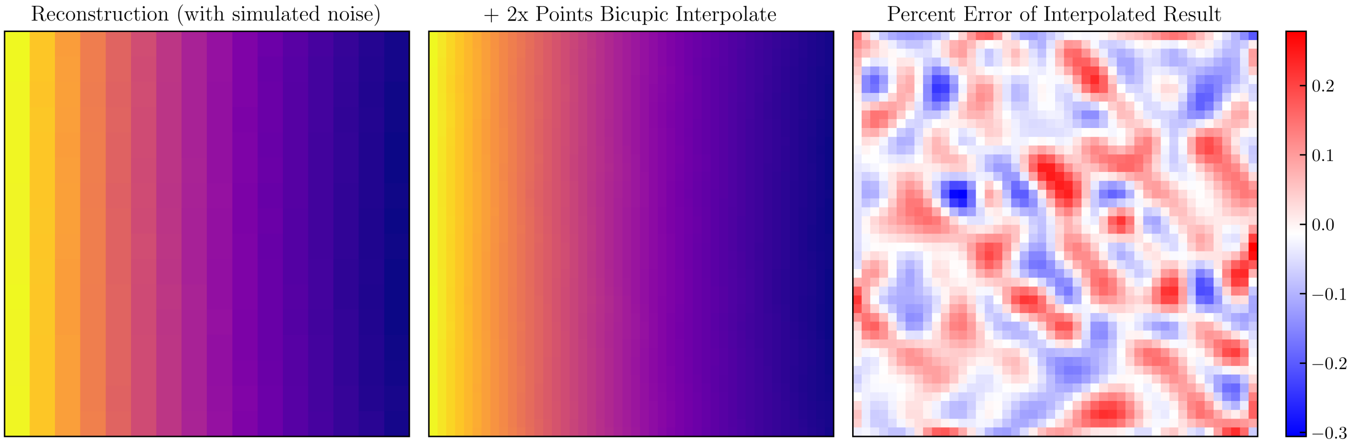

Fig. 6. Effects of bicubic interpolation, and error of interpolated result of reconstructed images, sharing a reconstruction model with Fig. 9.

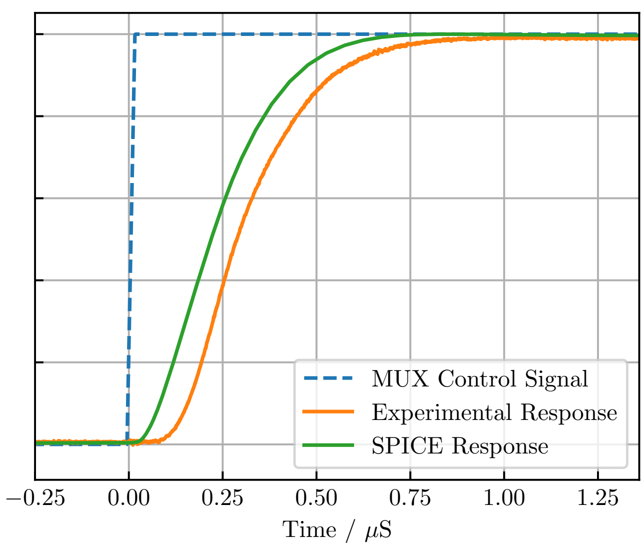

Fig. 7. Normalized experimental and simulated (with LTspice) filter settling behavior. The LTspice model is also shown in Fig. 11, and the experimental data was captured simultaneously with that of Fig. 15.

Sensor outputs are routed through high-performance analog multiplexers, buffered by low-noise rail-to-rail op-amps, and fed through single-stage Sallen-Key low-pass filters, before being buffered again by an ADC driver and digitized. The filter schematic is shown in Fig. 11, with an LTspice simulation of the filter response compared with the experimental settling shown in Fig. 7. The filter's pole frequency, $f_p \approx 1.2$ MHz, is tuned to allow five time constants of settling at a sampling rate of 40 kSPS, after accounting for the increased sampling rate required for multiplexing. This filter mitigates high-frequency switching noise and RF interference but does not serve as an anti-aliasing filter. Consequently, the system is susceptible to aliasing from interference in the 20–1,000 kHz range, but most sources of external noise (including the ISM band) should be sufficiently suppressed. The inability to implement effective anti-aliasing without incurring significant cost is the primary limitation of the multiplexed topology.

The chosen ADC is a 12-bit, simultaneously sampling, successive-approximation register (SAR) part, featuring four ADCs on a shared silicon multiplexed to eight inputs with a shared digital controller. These ADCs share a quad-SPI bus capable of clocking out samples as fast as 4 MSPS per ADC (for 16 MSPS total) [20]. To achieve the Nyquist frequency across the multiplexing regume, the ADC must sample at least 40 kSPS $\times 16 = 640$ kSPS per input channel, and the selected part has a maximum sampling rate of 1 MSPS per channel using the sequencer function, allowing a maximum sampling rate of just above $1.5\times$ Nyquist, which should be acceptable for real-world reconstructions [21].

In addition to the aforementioned common-mode immunity provided by the reference voltage design, both the ADCs and ADC drivers are shielded from potential interference inside μ-metal enclosures to prevent the ingress of noise, and significant work has been put in to best-practice PCB design approaches. The use of resistors to prevent ringing, multiple stitched ground planes, digital and analog traces intersecting across layers at 90° and a liberal deployment of LC filters in the power supply circuitry all contribute to reducing the generation and ingress of EMI.

B. Digital Electronics

The compact dimensions inside SOUTH's vacuum vessel necessitate a separate digital control assembly. The main analog PCB includes three buffer/level shifter ICs which enable operation with both 3.3 V and 1.6 V logic, allowing interfacing with a wide range of controllers over a standard, high-performance SYZYGY FPGA connector. So far two compatible digital interfaces have been designed: one for user-friendly, interactive operation using a microcontroller, and another for high-performance data acquisition using an FPGA.

The microcontroller-based solution uses a Teensy 4.1 [22] and is designed for sampling at up to 6 kSPS using dual hardware SPI interfaces. The Teensy can support 24 MHz one-wire SPI, but can achieve 48 MHz with reduced stability. In this configuration, the ADC's built-in hardware oversampling can be used to reduce bandwidth and mitigate aliasing, while also decreasing the amount of data produced. Full-rate sampling is not feasible due to bus speed and memory limitations.

To ensure data can be communicated correctly between the microcontroller and PC analysis software, a proprietary datapacket has been developed, with a layout as shown in Tab. I. This packet includes many synchronous bits at the start and end of the binary data, to make separating packets easier with inbluilt python and MATLAB functions like `serial.readuntil`. The packet includes empty space (the packet size padded to 512 bytes) and a version number to allow future expansion, and includes a packet counter and space reserved for CRC checksums to prevent both packet loss and data corruption. An example of the microcontroller digital logic operation is shown in Fig. 15 in the results section, and a representative sample of the microcontroller code is contained in Lst. 1 in the Appendix.

TABLE I
STRUCTURE OF SERIAL PACKETS

Field Name	Type	Bytes	Description
`sync1`	`uint8_t`	1	Sync byte 1 (0xAA)
`sync2`	`uint8_t`	1	Sync byte 2 (0xBB)
`sync3`	`uint8_t`	1	Sync byte 3 (0xCC)
`sync4`	`uint8_t`	1	Sync byte 4 (0xDD)
`count`	`uint32_t`	4	Packet counter
`timeStamp`	`uint32_t`	4	Optional timestamp
`dataSize`	`uint16_t`	2	Number of `uint16_ts`
`version`	`uint8_t`	1	Packet version (0x01)
`flags`	`uint8_t`	1	Reserved/unused flags
`data`	`uint16_t[192]`	384	Data payload
`spareData`	`uint8_t[104]`	104	Reserved
`CRC`	`uint32_t`	4	Space for Checksum
`sync5`	`uint8_t`	1	Sync byte 5 (0xEE)
`sync6`	`uint8_t`	1	Sync byte 6 (0xFF)
`sync7`	`uint8_t`	1	Sync byte 7 (0x11)
`sync8`	`uint8_t`	1	Sync byte 8 (0x22)
Total		**512**	Total packet size in bytes

The FPGA implementation uses a Pynq Z2 FPGA development board, capable of exceeding the 40 kSPS Nyquist frequency required for the 20 kHz sensor bandwidth. This is achieved using two four-wire SPI interfaces in parallel at bus speeds exceeding 50 MHz. The FPGA configuration is intended for high-performance, triggered measurements, though real-time interaction has been demonstrated as possible by other groups [17] and could be developed in the future.

C. Power Supply

Due to the analog electronics' reliance on the power rail for both a supply and reference voltages, a low-noise linear dropout regulator (LDO) is used to provide a stable and clean 3.3 V common carrier voltage from an input voltage in the range of 4–5 V. PSice simulations of this power supply, shown in Fig. 8, show acceptable startup behavior that meets the requirements set by the ADC [20], [23]. MOLEX micro-fit cables were chosen because of their extremely small PCB footprint, robust supply chain and adjacency with the mini-fit JR line of products already in use by SOUTH, with crimping tools already available in various accessible research labs. Ideally the device should receive its 4–5 V input voltage from an external linear power supply, but low-noise operation (as low as $4.4\,\mu V_{RMS}$ [23]) is still achievable when powered from a quality switching mode converter. The LDO is connected to several ground planes to ensure adequate cooling, but running on closer to 4 V_{in} is advisable to prevent excessive heat dissipation.

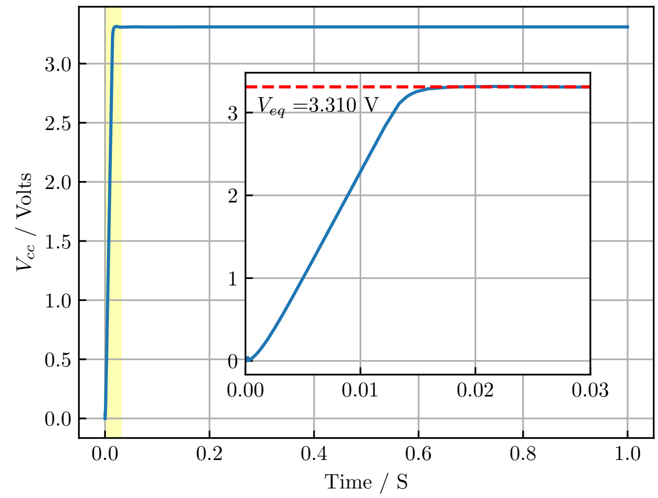

Fig. 8. PScipe simulation of power supply output voltage, with detail showing startup time. In steady-state operation noise is expected to be $4.4\,\mu V_{RMS}$ [23].

D. Data Processing & Analysis

To convert the raw ADC output codes into meaningful images, each code is mapped to a spatial location in two dimensions, then converted to a voltage and a magnetic field using the sensor sensitivity and DC offset. This processing is performed externally on a host computer. For the FPGA-based implementation, additional processing is required to apply a phase correction to each sensor's fully reconstructed (with $\sin x/x$ interpolation) analog waveforms to correct for rolling-shutter effects that are an inevitable consequence of the multiplexed topology. For real-time operations, the acquisition rate is significantly faster than the display refresh rate, rendering rolling shutter effects negligible.

Bicubic interpolation between adjacent sensors has been employed in previous Hall array implementations [14], [17], and is used here in post-processing where the computational cost is negligible. Fig. 14 presents selected results enhanced via bicubic interpolation; this method is effective when the spatial resolution of the sensor array exceeds the granularity of the magnetic field being reconstructed, such as in the case of the tokamak's toroidal field cross-section, but localized field—such as those generated by small button magnets—are vulnerable to artifacting.

E. Intended Use in SOUTH Tokamak

The primary application of the device in the SOUTH tokamak is to validate the magnitude and spatial configuration of the toroidal field, which should ideally look like the field shown in Fig. 2. Fig. 9 illustrates a numerically simulated

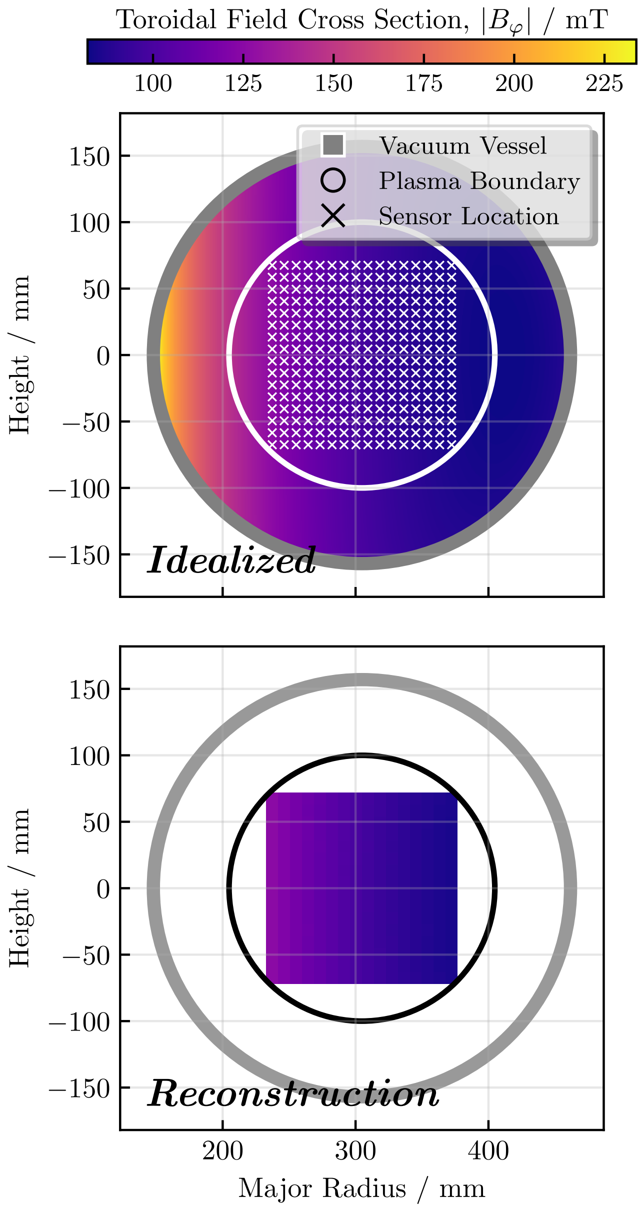

Fig. 9. Idealized plot shows sensor geometry (a 9 mm pitch) within SOUTH and an ideal field simulation during the ‘flat top‘ of a 3 kA pulse, shared with Fig. 2; the reconstructed field accounts for sensor geometry and noise.

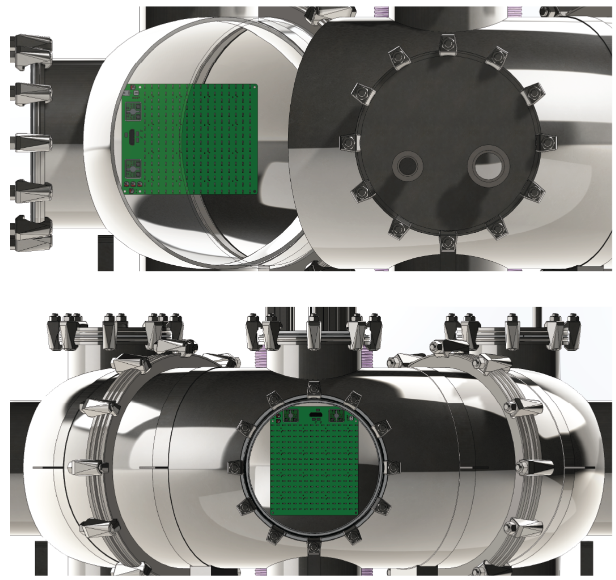

Fig. 10. Positioning of the tile within SOUTH. The top configuration monitors toroidal magnetic field while the bottom configuration monitors radial component. The device could also be positioned facing upwards to measure the poloidal field.

toroidal field for a 3 kA plasma in SOUTH, alongside a simulated reconstruction incorporating modeled device noise. The imaging capability of the array is expected to confirm the characteristic $B_\varphi \sim 1/R$ relationship and allow comparison with multiphysics simulations. The device could also potentially offer some quantitative measurement of magnetic field ripple; a device bandwidth of 20 kHz will be able to get some basic measurements of the ripple in the 10 kHz switching frequency device, but because that 10 kHz signal is a square wave a full reconstruction would not be possible (and would require a sampling rate of at least 30–50 kSPS by rule of thumb [21]).

While applying this device to the full assembled tokamak system (in the configuration shown in the upper segment of Fig. 10) is the primary design goal, the device will also serve a critical role in the research and development of the individual coil and power supply systems before the full design integration. Preliminary tests of toroidal field coils at fractional currents are expected to begin by the end of 2025, and this device will be able to verify that the applied currents are achieving the desired field strengths. The instant feedback that the Hall array is able to provide will be able to improve the efficiency of testing by replacing the use of a single, lower-bandwidth gauss meter, and allow for the timely detection of any manufacturing or assembly defects.

In addition to the toroidal field, the device can also be applied to the monitoring of the radial magnetic field, which should ideally be zero at the center of the toroidal field. Deviations from this expectation would indicate the presence of stray magnetic fields, and this would in turn indicate issues with the magnetic field system that could be flagged for further investigation. Since each sensor measures the field component parallel to its normal vector, the array must be positioned to intersect the field lines of interest, as shown in the bottom section of Fig. 10. A poloidal application, similar to [9], is also possible, but this would be particularly intolerant of sensor misalignment and require a high sensitivity.

F. Initial 1/16 Scale Prototype

Because of the significant cost investment of the final device (each sensor cost about A$ 1 and each ADC about A$ 80, with a minimum order quantity of two devices) a 1/16th scale prototype was designed to test the PCB manufacturing quality and analog electronics. The prototype, shown in Fig. 11, featured a two-stage analog filter that was unstable in its manufactured state. After making some modifications to the filter and reducing it to a single stage, the filter achieved the performance shown in Fig. 11.

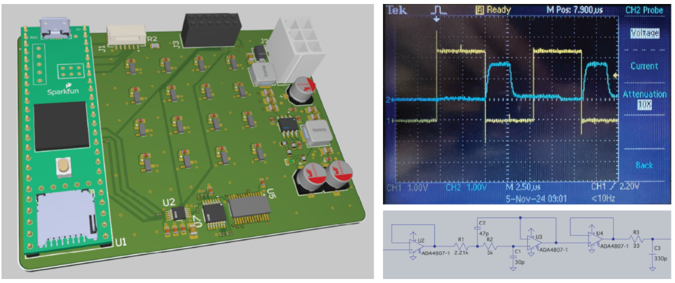

Fig. 11. 1/16th Scale Prototype, with experimental filter testing results and filter schematic shown.

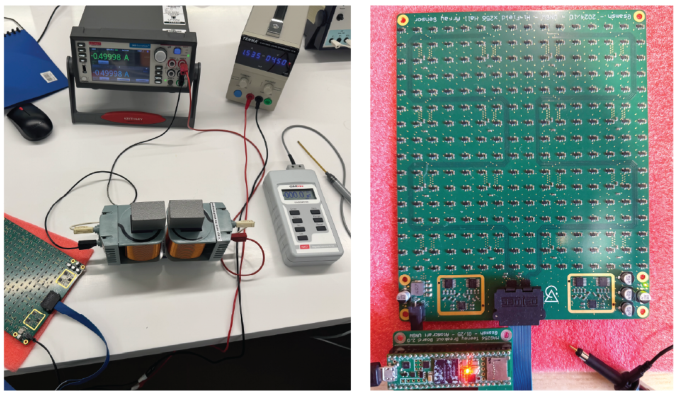

Fig. 12. Linearity calibration setup with source meter, gauss meter, and ferrite core electromagnet (left), and close-up of the completed device (right) with the Teensy digital controller board partially shown.

The prototype used `0201` size components, initially chosen so that a full two-stage filter setup could be fit inside a surface-mount EMI shield, but the difficulty in modifying the components (because they were so small) combined with the dropping the second filtering stage informed a switch to much more accessible `0603` components for the final design. This decision was also influenced by the significant cost penalty that `0201` assembly incurred, and the increased parts availability (i.e., a wider range of resistances and capacitances to use a more favorable filter configuration) and flexibility provided by `0603` parts that could be soldered by hand to accommodate changes to the filters or drivers after initial manufacturing.

Another key objective of the prototype was testing the manufacturing tolerances of the 'via-in-pad' technology, which dramatically reduces routing complexity by embedding vias directly into the very fine pitch (< 0.65 mm) pads of the multiplexers and op-amps. No short circuits between adjacent pads were identified by the manufacturer, and none were found during manual inspection. The prototype also functioned as a test-bed for the development of PC-based analysis software while the final device was still having its design finalized, parts sourced and circuit boards printed, dramatically accelerating the project's software development timeline.

III. Results

The final device (shown in Fig. 12) was delivered with no debilitating design of manufacturing defects, and testing to-date is indicative of a comprehensive fulfillment of the design goals outlined in the introduction. The micocontroller digital control board is capable of much faster sampling than anticipated, with sampling up to 10 kSPS possible. This makes further development of the FPGA implementation unnecessary until the full 40 kSPS sampling is required for full-bandwidth testing of SOUTH's coils, anticipated for mid–late 2026. Interactive plotting in real-time also works reliably, although software Python and MATLAB-based serial frontends limit the data collection rate to about 250 SPS, meaning 10 kSPS sampling can only be achieved over short periods of time when taking a 'triggered' measurement. This means that an FPGA development board—with external, discrete and plentiful DRAM—is necessary for extended operation at high sampling rates.

Some preliminary static imagery (averaged over many acquisitions) is shown in Fig. 14, but the device's functionality and dynamic imaging capabilities are best demonstrated more interactively as in the video contained within the supporting material or hosted on YouTube, here. This video should be considered **essential viewing**.

A. Quantification of Noise

The noise of all individual sensors were measured with no applied magnetic field over a sampling period of 10,000 samples, collected with a generous settling time to maximize the effectiveness of the analog electronics. As shown in Fig. 13, all 256 sensors perform similarly and there is no significant geometric correlation of noise statistics between adjacent sensors or sensors that share filters or ADCs. The device demonstrates a very low AC RMS noise floor below one least-significant bit or about 0.086 mT, with a corresponding effective sensitivity of 0.32 mT. The noise statistics do not drop off as expected with the introduction of hardware oversampling, likely because the noise floor is dominated by quantization noise. This indicates that future iterations could benefit from the use of higher-resolution ADCs, although the utility of such an improvement is questionable.

This is an exceptionally low noise floor, well in excess of the requirements identified as part of the projects literature review. This noise floor is low enough that it likely enables new uses of the device, specifically in the monitoring of SOUTH's (much smaller) poloidal fields. The high dynamic range of the instrument, combined with its sensitivity, would allow for the monitoring of small amounts of ripple ontop of large DC magnetic fields.

B. Calibration

The DC offset at zero magnetic field was calculated using the same dataset as the noise measurements. Sensor calibration was performed with a ferrite-core electromagnet powered by a Tektronix/Keithley 2460 source-measure unit, generating a highly stable and uniform magnetic field. This field was measured with a four-digit gauss meter to be approximately 115 mT, with cross-sectional variations under 1%. Each sensor on the PCB recorded both positive and negative field measurements (i.e., aligned and anti-aligned to the same field). These high and low-field data points, along with the zero-field

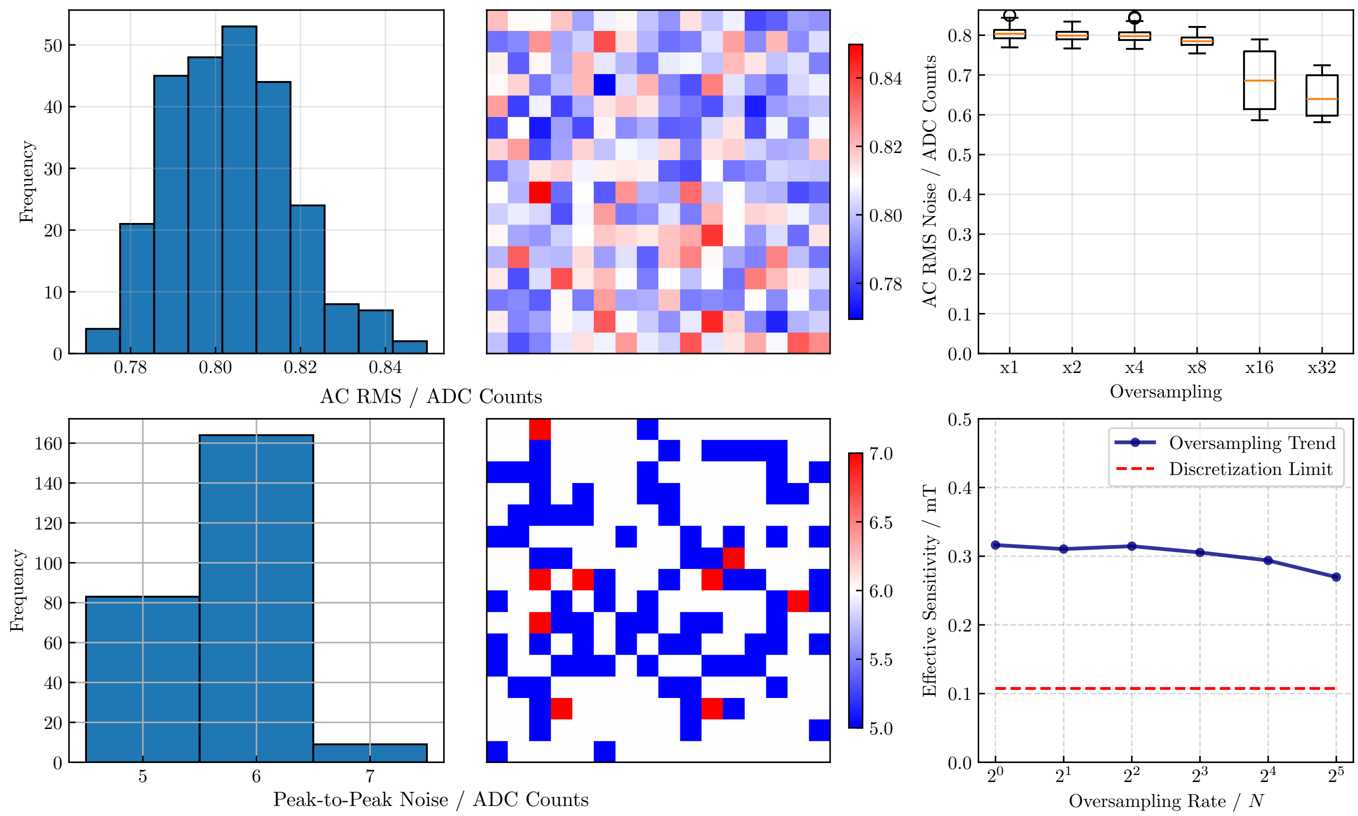

Fig. 13. Peak-to-peak and AC RMS noise data and spatial distributions, AC RMS noise boxplots as a function of hardware oversampling window size, and effective sensitivity as a function of hardware oversampling window size.

average, were binned into histograms and used to perform a three-point linear regression, with the resulting sensitivities shown in Fig. 14. This calibration focused on determining the linear sensitivity of each sensor, and not validating the sensor linearity that was already guaranteed by the manufacturer [19].

Both the linearity and DC offset constant show no significant or persistent geometric correlation across the sensor plane, which would be indicative of a misbehaving filter or some source of localized noise on the PCB, as shown in Fig. 14. While sensitive enough to meet the requirements set for the device, observed sensitivities were still generally below the quoted values from the sensors' manufacturer [19], potentially indicating that the filters have a gain slightly but consistently higher than unity, which could be explained by capacitors generally erring on the side of being bellow the stated capacitance rather than above (although still within the quoted 5% tolerances).

C. Experimental Performance of Electronics

Using an oscilloscope with a logic analyzer add on, a full digital and analog capture of an acquisition event can be made simultaneously. Fig. 15 shows the full digital signal capture of the digital logic going to and from the micro-controller, running at a reduced sampling rate. This logic proves that everything is working as expected, and highlights where the parallelization provided by an FPGA could be used to dramatically increase performance. Timings could also be tightened to achieve higher a sampling rate by accounting for the time taken to move SPI results from the input buffer to memory, which would reduce the required programmed wait times between transactions.

Similarly, Fig. 7 shows the experimental filter response of the final filter design, slightly improved over the prototype shown in Fig. 11. The filter settles slightly faster here with only a single Sallen-Key stage, at the cost of a more relaxed attenuation response in the frequency domain. The experimental filter is very close but still noticeably slower in settling than the LTspice simulation, which is likely due to either an overly optimistic spice model provided by the manufacture of the op-amp and/or multiplexer (either an overstated slew time or underquoted switching speed), or from the spice model not adequately modeling the effects of the ADC on the filter circuit (a lack of modeling of input capacitance, for example). Future iterations of the device should raise the pole frequency, f_p, to combat this slower than expected performance.

IV. Discussion

The scope of the 2D Hall array in this configuration is inherently limited to small, spherical tokamaks where the array dimensions can be reasonably compared to the toroidal region of interest while maintaining sufficient sensor granularity. This device offers small fusion endeavors a cost-effective alternative to the comprehensive field mapping solutions available to larger, more mature projects.

The device demonstrates significant pedagogical potential, serving as digital version of magnetic viewing film. It could

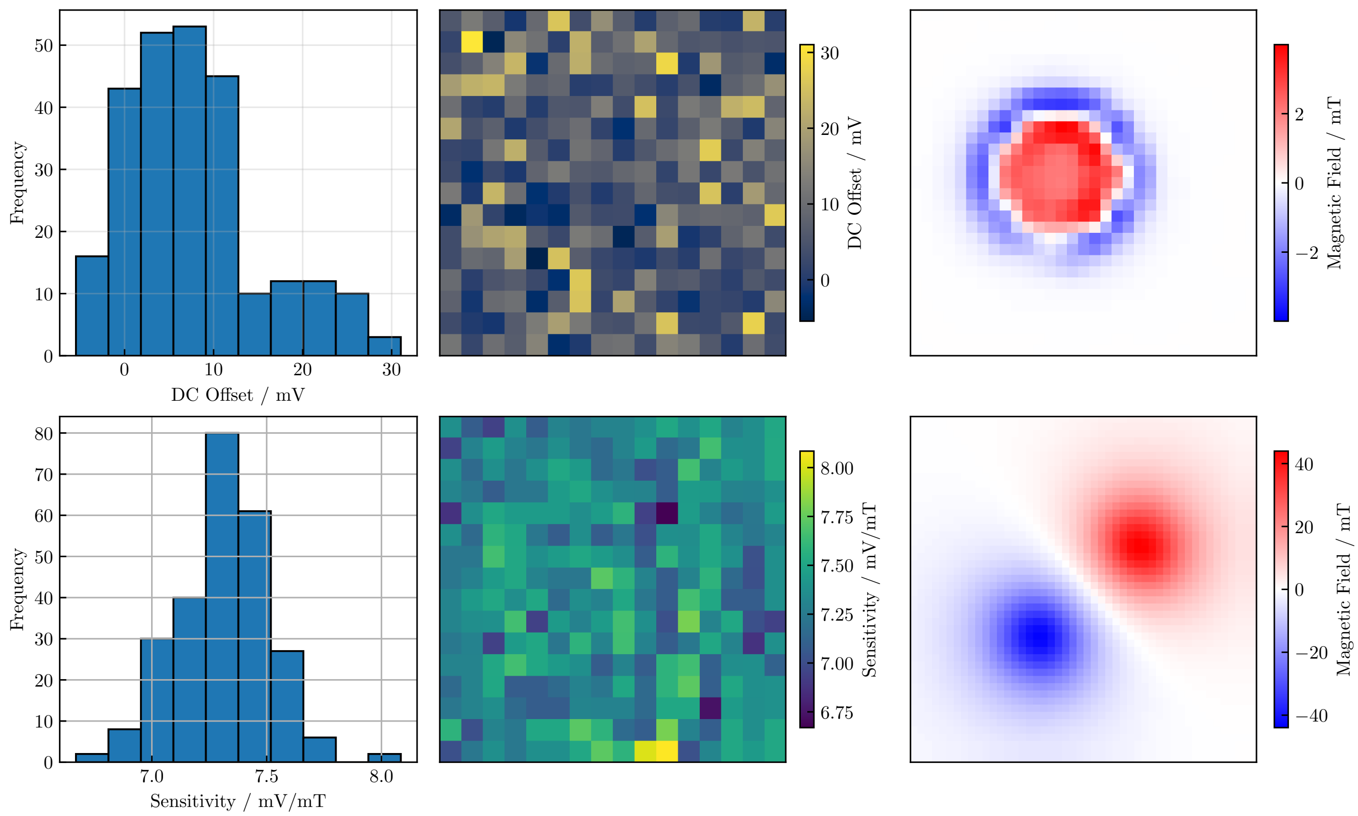

Fig. 14. Calibration results – showing the DC offset of a zero field (top) and the linear magnetic field sensitivity (bottom) – and preliminary imagery (right) of an Apple MagSafe charger and a very strong bar magnet, showing the use of bicubic interpolation. The manufacturer quotes an expected sensitivity of 7.5 mV/mT [19].

enhance undergraduate instruction in general physics and introductory electromagnetism courses by enabling direct observation of fundamental concepts, like the magnetic fields produced by current-carrying wires. The visual and interactive nature of the device makes it particularly suitable for outreach activities, and *AtomCraft* will use it to attract new students recruits to the project and the University broadly, with the device expected to be presented to the general public at the upcoming UNSW Open Day in early September 2025.

The array's sensitivity, dynamic range and low noise floor open possibilities for its use as a plasma diagnostic tool in SOUTH for monitoring poloidal fields (where the intrusion into the plasma is not as prohibitive to tokamak operations), as demonstrated previously [9]. However, its application in larger tokamaks like ITER or commercial reactors faces challenges. In addition to the aforementioned geometric limitations, its reliance on commercial sensors restricts operation in high-radiation environments, a problem avoided by ITER's custom bismuth sensors [6], for example. Additionally, the multiplexed topology introduces aliasing risks for high-frequency plasma instabilities above 20 kHz. Future iterations could explore custom sensors or more exotic analog filtering and ADC configurations to address these limitations, potentially expanding the device's utility to both engineering validation and plasma physics in small-to-medium size fusion devices.

A. Immediate Future Developments

The immediate priority for future development of this work is the continued design of an FPGA digital control board which can 'slot in' to replace the Teensy microcontroller. A breakout board to adapt the connector that comes out of the device to interface with a Pynq Z2 FPGA development board has already been designed, manufactured and assembled, and some preliminary FPGA development has already commenced. The better than expected performance of the microcontroller implementation have lessened the requirement to have the FPGA ready as soon as possible, but it is still required to take full advantage of the bandwidth on offer and is therefore still a future goal.

B. Presentation at SOFE 2025

While presenting this work at the Symposium on Fusion Engineering 2025, on the MIT Campus in Cambridge, MA, there were a few attendees who had suggestions or observations about potential applications of the device, these are included here as potential future developments.

One attendee from the MIT Plasma Science and Fusion Center (PSFC) asked about the prospects of using a similar device submerged in liquid nitrogen for the monitoring of superconducting magnets, within the context of MIT PSFC's new superconducting magnet test facility [24] which is being used for the development of new tokamaks and stellarators

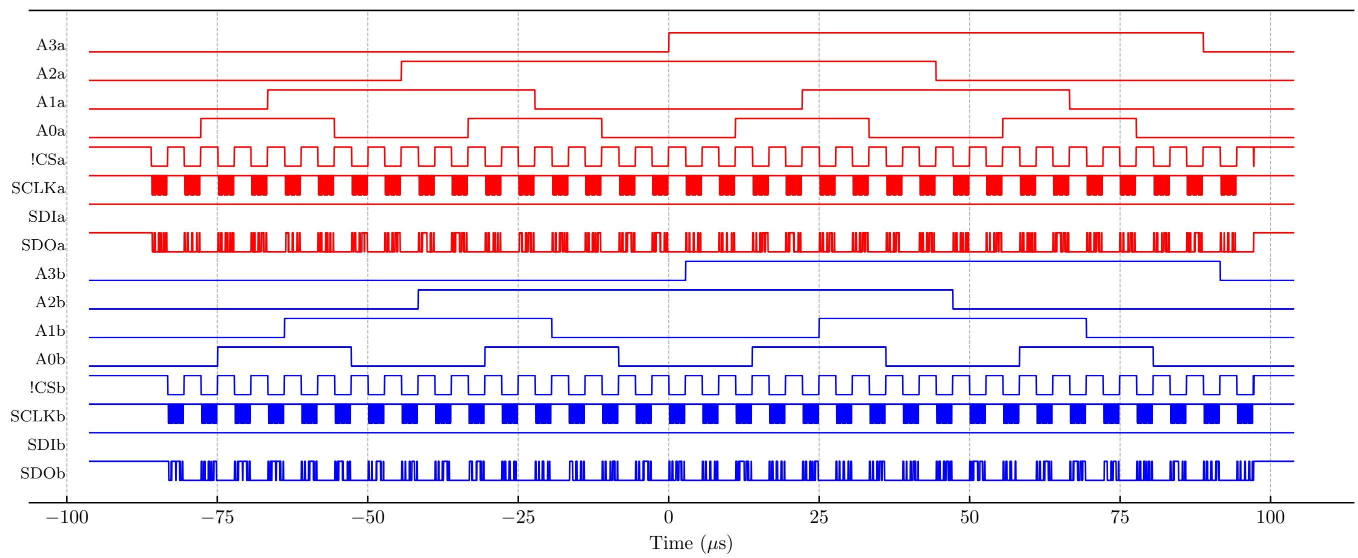

Fig. 15. Experimental capture of the digital logic behind a single 'image' taken using the digital channels of an oscilloscope (this data was recorded at the exact same time as that of Fig. 7). Compare this with the code shown in Listing 1 in the supporting material to get a better idea of what each SPI transaction is for. Note the very quick 'reset sequencer' pulse on the chip select lines towards the very end of the data.

like Commonwealth Fusion Systems' (SP)ARC reactors [25] and Type One Energy's FPP [26].

This application is extremely novel and innovative, but equally challenging and technically demanding. The device would need to be redesigned so that only the Hall sensors—and not any of the supporting electronics—get dipped into the liquid nitrogen, with all analog processing and data acquisition happening externally, and different, high-field Hall sensors would also need to be used. Getting the Hall sensors to function while submerged in liquid nitrogen initially seems like a technical hurdle, with the datasheet discouraging operation outside of a quite restrictive thermal envelope [19], but another attendee from OpenStar technologies (a New Zealand-based levitating dipole fusion company [27] that *AtomCraft* collaborates with) remarked that they semi-regularly use off-the-shelf Hall effect sensors in similar conditions with full functionality, even when operating well outside of datasheet specifications.

Another attendee from the Princeton Plasma Physics Laboratory (PPPL) asked about the possibility of using a similar array for liquid metal tomography and identified an extremely high dynamic range as critical for this application, with the ability to distinguish small fields (mT range) in the pretense of a very large background (multiple T) the key requirement. An array of Hall sensors is suited to this task, but likely custom sensors would be needed, potentially with a tuned analog bias current applied to cancel out the passive magnet field, as the dynamic range requirements are otherwise so extreme.

C. Future Collaboration with SMART Tokamak in Seville

After networking and exchanging contact details at SOFE 2025, a preliminary collaboration between these authors and the SMART Tokamak team—a collaboration between the University of Seville and the PPPL [4]—has begun with the goal of using a small set of similar devices to monitor and control the alignment of coils during the re-assembly of the SMART tokamak, which is happening until late October 2025.

The preliminary concept is to apply a 'small' DC currents to the tokamak's magnetic coils, and then use a set of about six arrays in key locations to confirm that all the fields are adding as constructively as possible as each coil is moved into position and secured. The first author is expected to visit SMART in Seville in late 2025 to assist with this work, and communication is ongoing between the two teams currently.

V. Conclusion

This paper has presented the design, performance, and preliminary results of a high-resolution 2D Hall effect array device built to aid in the research and development of a small spherical tokamak. By functioning as a magnetic field camera, the arrays can image the toroidal fields within the region of plasma confinement and de-risk engineering efforts by verifying that generated magnetic fields match multiphysics simulations.

The device demonstrates a high effective resolution of 0.32 mT, an average AC RMS noise floor of 0.086 mT and a dynamic range exceeding ± 150 mT. The completed system provides a practical, cost-effective solution for magnetic field mapping in resource-constrained environments while offering substantial educational value. The low noise performance achieved with commercial components establishes a new benchmark for future developments of similar systems, in fusion engineering and beyond.

Following a positive reception from the fusion community at SOFE2025, this device will be used in the assembly of SOUTH in the medium-term future—up until late 2026—and will be applied shortly to the assembly and commissioning of the SMART tokamak, in preparation for its second experimental campaign envisioned to commence in late 2025/early 2026.

Acknowledgments

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He has been involved in the *AtomCraft* project since May 2024 and has served as diagnostics team lead, designing the diagnostics systems architecture for the upcoming SOUTH tokamak. He has extensive undergraduate research experience in both mechanical engineering and physics, in both industrial and academic settings.

He was been awarded the UNSW Nuclear Innovation Centre and Tyree Foundation travel grant, as well as the Gates Cambridge Scholarship for doctoral study commencing in October 2025.

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From 2013 to 2014, he was a Research Associate at Cambridge University. Then, he worked at Mitsubishi Heavy Industries, Japan, from 2015 to 2017. Since 2022, he has been a Senior Lecturer in robotics and mechatronics at UNSW Sydney, Australia. His previous work includes optimal control of electric machines to reduce energy consumption and torque ripple as part of the Virtual Vehicle Integration and Development (ViVID) project and optimization of electrification of rail routes to reduce carbon dioxide emissions and energy usage. His current work focuses on applying mechatronics, control engineering, and machine learning to transportation systems to reduce energy consumption and carbon dioxide emissions.

Dr Midley is the mechatronics academic lead for *AtomCraft*.

Supplementary Materials

A. Videos of Device in Operation

Essential Viewing: View the device in operation on YouTube https://youtu.be/sB7mTB_tMe0.

B. Schematics & Other Files

The full device schematics and Altium PCB files are hosted on GitHub at https://github.com/Sam-js2/BEThesis, as well as the poster that was presented at SOFE 2025 and the Thesis C conference presentation.

C. Microcontroller Code

The full code for the real-time interaction module (shown in the supporting video) can be found on GitHub at https://github.com/Sam-js2/BEThesis. Some code is shown here, but some functions are only defined in the full codebase.

```cpp
void setup() {

  Serial.begin(115200);

  SPI.begin();
  SPI.beginTransaction(SPISettings(spiClock, MSBFIRST, SPI_MODE2));
  SPI1.begin();
  SPI1.beginTransaction(SPISettings(spiClock, MSBFIRST, SPI_MODE2));

  delayMicroseconds(t_powerup) ;

  adcA.hardReset();
  adcB.hardReset();

  adcA.configADC();
  adcB.configADC();

  delayMicroseconds(cycleWait);
}

void loop() {

  // Junk data out, these trigger measurements
  adcA.spiRead()                                  ;
  adcB.spiRead()                                  ;
  delayNanoseconds(wait1)                         ;

  for (int i = 0; i <= 15; i++) {

    // Read THEN multiplex
    adcA.spiRead(&txPacket.data[6*i + 0 ])       ;
    adcA.setMUX(gray[i+1])                       ;
    adcB.spiRead(&txPacket.data[6*i + 96])       ;
    adcB.setMUX(gray[i+1])                       ;

    delayNanoseconds(wait2) ; // Filter settling

    adcA.spiRead(&txPacket.data[6*i + 3 ])       ;
    adcB.spiRead(&txPacket.data[6*i + 99])       ;
    // Pointers direct data to correct place in packet strut
  }

  transmit()                                      ;
  pulseBothCS() ; // Resets Sequencer
  delayMicroseconds(cycleWait) ; // Spacing between captures
}
```

Listing 1. C++ setup function and main loop for real-time operation on the Teensy 4.1.